Amazon Product Manager INSIGHT ENGINE

Qualitative + Quantitative Intelligence Platform for Amazon PMs

Product Requirements Document
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Product	Status	Target Users
PM Insight Engine	**Draft PRD**	Amazon L5–L7 PMs

1. Feature

Feature Name: PM Insight Engine: Unified Qualitative + Quantitative Intelligence Dashboard
Product Area: Product Discovery & Opportunity Identification
Target Platform: Web application (desktop-first)
Prototype Stack: n8n (workflow orchestration) + Claude (Frontend Website) + Gemini 2.5 Flash Lite + Google Drive

> *PM Insight Engine is an AI-powered synthesis and dashboard platform that ingests fragmented qualitative customer signals (reviews, tickets, NPS, transcripts, chat logs) and quantitative business data (KPI exports, analyst SQL outputs, A/B results, Excel data) into a single unified, scored, filterable intelligence view, enabling Amazon PMs to identify and prioritise the highest-impact customer problems in under 60 minutes instead of 3–5 days.*

2. Core Problem

The Structural Intelligence Gap

Amazon PMs cannot systematically combine qualitative customer signal with quantitative business metrics into a single, scored, filterable intelligence view thus forcing them to spend 3–5 days per discovery cycle on manual synthesis that produces incomplete, un-scored, and untraceable insight.

Details:
Amazon Product Managers operate at a scale where millions of qualitative signals like customer reviews, support tickets, NPS comments, interview transcripts, chat logs that exists simultaneously alongside dense quantitative data: A/B experiment outputs, KPI dashboards, analyst SQL reports, and Tableau exports.

These two data types live in separate systems, interpreted by separate teams, on separate timelines. The synthesis layer between them does not exist. PMs bridge the gap manually to a process that takes 3–5 days per discovery cycle, produces biased and incomplete output, and leaves the PM with a narrative document rather than a structured, scored, traceable artifact.

Five Structural Breakdowns

#	Breakdown	Root Cause	Impact on PM	Severity
1	Qualitative–Quantitative Split	Reviews/tickets in Voice of Customer (VoC) tools; metrics in Redshift/Tableau. No shared schema.	Roadmap decisions made on one signal type at a time. Incomplete picture.	HIGH

2	WBR (WEEKLY BUSINESS REVIEW) Prep Consumes Strategic Days	PM spends Mon–Tue manually assembling metric narrative. No auto-correlation with VoC.	2 full strategic days lost weekly. PM reports on the past, not the future.	**HIGH**
3	Escalations Treated as Isolated Events	No clustering of complaints. Each escalation triggers a full-day investigation.	Roadmap interrupted. Same root cause investigated repeatedly.	**HIGH**
4	Qualitative Has No Quantitative Amplification	UX interview findings live in Google Docs. Never scored, linked to metrics, or routed.	High-value insights deprioritised because they cannot be justified with numbers.	**MEDIUM**
5	No Traceable Output Artifact	Insight synthesis lives in ad-hoc documents. Cannot be filtered, queried, or exported.	Each PRFAQ (PRESS RELEASE FAQ) or WBR (WEEKLY BUSINESS REVIEW) starts from scratch. Institutional insight is lost.	**MEDIUM**

Why Current Solutions in market are Insufficient

Tool	What It Does	Why It Fails Amazon PMs
Dovetail / EnjoyHQ	Qualitative research repository	Qual-only. No quantitative integration. No Amazon-specific routing or PRFAQ (PRESS RELEASE FAQ) output.
Amplitude / Mixpanel	Quantitative product analytics	Quant-only. Cannot process text, reviews, or interview transcripts.
Tableau / QuickSight	BI dashboards	Displays data already structured. Cannot synthesise unstructured qualitative signal.
Confluence / Notion	Document storage	No AI processing. No scoring. Insight lives in docs, not queryable systems.
Productboard	Feature prioritisation	Requires manual insight entry. No AI synthesis. No Amazon-specific workflow fit.

Why This Problem Now

- LLMs can now cluster and score qualitative text at production quality. This was not reliably possible 18 months ago.
- Amazon's WBR (WEEKLY BUSINESS REVIEW) culture demands narrative + numbers together. The tool closes that gap structurally.
- PRFAQ (PRESS RELEASE FAQ) process requires a validated customer problem before any build. Manual synthesis is the bottleneck to that validation.
- Amazon PMs already have all the raw data. The missing layer is synthesis and unification and not data collection.

3. Users & Their Needs

Primary Users

The Analytical Discoverer , Amazon L5/L6 Product Manager

Attribute	Detail
Role	L5 or L6 PM owning a product surface or category (e.g., checkout, search, returns, Seller Central onboarding)
Experience	3–7 years PM experience; 1–3 years at Amazon or equivalent data-heavy environment
Team Context	Works with 1–2 shared data analysts, 1 shared UX researcher, engineering counterpart
Weekly Data Exposure	WBR (WEEKLY BUSINESS REVIEW) metrics, A/B experiment results, VoC summaries, escalation emails, analyst SQL outputs, UX session notes
Primary Tool Stack	Redshift/Athena (SQL), Tableau, Confluence, internal VoC dashboard, JIRA
Key Deliverables	Weekly WBR (WEEKLY BUSINESS REVIEW) narrative, PRFAQ (PRESS RELEASE FAQs), 6-pager documents, backlog prioritisation, escalation responses
Trust Profile	Deeply sceptical of AI hallucination. Will reject output that cannot be traced to a verbatim source.
Adoption Trigger	Has 3 days to prepare a PRFAQ (PRESS RELEASE FAQ). Has 400 reviews, 2 analyst reports, 8 UX notes, and 50 support tickets.

Current Pain Points, Primary User

- Spends 6–12 hours per week manually synthesising qualitative signal into summary documents.
- Has no consistent scoring methodology, theme priority based on gut feel and recency bias.
- Cannot connect a metric drop to a customer reason without running separate ad-hoc analysis.
- PRFAQ (PRESS RELEASE FAQ) customer problem section is the hardest, most time-consuming section to write credibly.
- Escalations interrupt the roadmap because there is no system to detect that the same complaint has occurred before.
- Qualitative insight lives in Google Docs, cannot be filtered, queried, or reused across cycles.

Jobs-to-be-Done , Primary User

- JTBD 1: When I am preparing a PRFAQ (PRESS RELEASE FAQ), I need to identify the top 3 customer problems with evidence, so I can write a credible "customer problem" section without spending a full day on manual research.

- JTBD 2: When I am preparing for WBR (WEEKLY BUSINESS REVIEW), I need to understand why a metric moved, so I can explain the root cause to leadership with both numbers and customer voice.
- JTBD 3: When I receive an escalation, I need to know whether this is a one-off or a systemic pattern, so I can respond proportionately and triage it to the right team.
- JTBD 4: When I present to cross-functional stakeholders, I need a single document that shows qual and quant together, so the case for a feature is immediately understood without multiple source docs.

Secondary Users

The Strategic Reviewer , Sr. PM / PM-T (L7+)

Attribute	Detail
Role	Sr. PM or PM-T overseeing multiple product areas or leading a product family
Use Case	Reviews synthesised output from L5/L6 PMs. Needs cross-product pattern visibility.
Key Need	Filterable view across multiple product areas. Trend over time. Export for leadership review.
Pain Point	Currently reviews L5/L6 PM Google Docs that have no consistent structure or scoring methodology.
Adoption Driver	If the L5/L6 PM uses it and the output quality is high, the Sr. PM adopts it to review their work.
Secondary Need	Ability to compare Opportunity Scores across product areas to guide quarterly roadmap allocation.

The Routing Recipient , Amazon Department/Team Lead

Attribute	Detail
Role	Owner of the department or feature area to which a theme is routed (e.g., Checkout Team, Outbound Logistics)
Use Case	Receives structured theme summary with verbatims, frequency, and severity, not a raw complaint dump.
Key Need	Enough context to act without re-investigating from scratch.
Adoption Driver	Tool routes to them automatically. They receive a JIRA-ready ticket structure with source evidence.

4. Current State Analysis

4.1 How PMs Synthesize Insight Today

The following is the actual workflow a typical Amazon L5/L6 PM follows to prepare a PRFAQ (PRESS RELEASE FAQ) or WBR (WEEKLY BUSINESS REVIEW) narrative. Every step is manual, every tool is separate, and no output is reusable.

Step	Action	Tool Used	Time Cost
1	Export customer reviews to spreadsheet	Amazon VoC internal tool → Excel	30–60 min
2	Manually read and colour-code themes	Excel, no taxonomy	2–4 hours
3	Pull support ticket summaries from BI report	Internal BI dashboard → copy/paste	45 min
4	Run SQL query for metric context	Redshift / Athena	1–2 hours (if analyst available)
5	Export Tableau dashboard as screenshot	Tableau → PNG	15 min
6	Consolidate into a Google Doc	Google Docs, narrative bullets	1–2 hours
7	Estimate severity and frequency from memory	No system, gut feel	30 min
8	Draft PRFAQ (PRESS RELEASE FAQ) customer problem section	Blank Google Doc	1–2 hours
	TOTAL		**6–12 hours per cycle**

4.2 What Is Broken Structurally

- No shared schema: Reviews, tickets, NPS comments, and SQL outputs use different fields, formats, and terminology. Manual translation required at every step.
- No scoring methodology: Themes are ranked by PM intuition, not frequency or severity. Confirmation bias is structurally guaranteed.
- No source traceability: By the time a theme reaches a PRFAQ (PRESS RELEASE FAQ), the original customer quotes are often lost or paraphrased.
- No routing intelligence: A product quality complaint looks identical to a logistics complaint until a human read both carefully. No automatic triage to the right team.
- No reusability: Each synthesis cycle starts from scratch. Learning from the previous cycle is informal and personal, it doesn't live in a system.

5. Solution Definition: PM Insight Engine

5.1 What It Is

PM Insight Engine is a structured AI processing pipeline with an interactive dashboard front end. It ingests both qualitative and quantitative data, processes them through a six-stage AI synthesis layer, and produces a unified, scored, filterable intelligence view with drill-through access to verbatim evidence and one-click export to PDF or DOCX.

It is not a chatbot. It is not a general analytics platform. It is a deterministic workflow tool with AI at the synthesis and scoring layers, and structured output at the delivery layer.

5.2 Before and After

Before PM Insight Engine	After PM Insight Engine
6–12 hours of manual synthesis per cycle	**Under 60 minutes from upload to scored output**
Qualitative and quantitative always separate	**Both combined in one filterable view per theme**
Themes ranked by gut feel	**Themes ranked by Opportunity Score (frequency × severity)**
No automatic routing to team/product area	**AI classifies each theme to department and feature area**
PRFAQ (PRESS RELEASE FAQ) customer problem written from memory	**PRFAQ (PRESS RELEASE FAQ)-ready draft generated from top-ranked theme**
No verbatim traceability	**Every theme links to original source quotes**
No consistent export format	**One-click PDF/DOCX snapshot for leadership reviews**

5.3 Core Output Artifacts

- Opportunity Scorecard, ranked list of top 5 customer problem themes with score, risk level, affected segment, and trend.
- Theme Deep-Dive Card, drill-through view per theme showing frequency, sentiment breakdown, verbatim clusters, affected product area, recommended owner, and PRFAQ (PRESS RELEASE FAQ)-ready problem statement.
- Quantitative Context Panel, plain-language interpretation of analyst data linked to each theme, with metric movement summary and hypothesis implications.
- Exportable Intelligence Brief, filtered snapshot of the dashboard in PDF or DOCX format, preserving all scores, verbatims, and metric context.

6. Proposed Feature

High-Level Description

PM Insight Engine is a six-stage AI processing pipeline with an interactive dashboard front end. The PM uploads both qualitative and quantitative data files. The pipeline processes them automatically and delivers a unified, scored, filterable Opportunity Dashboard with drill-through access to verbatim evidence and one-click export to PDF or DOCX for leadership reviews.

Key Components

Component	Description	Output
Ingestion Engine	Accepts CSV/Excel for both qual and quant inputs. Normalises to unified schema.	Structured JSON rows ready for AI processing
AI Synthesis Pipeline	6-stage AI pipeline: extract themes → cluster → score → route → interpret quant → generate PRFAQ (PRESS RELEASE FAQ) draft.	Ranked Opportunity Scorecard with scoring and routing
Unified Dashboard	Interactive filterable view combining qual themes with quant metric context. Top-5 cards, trend view, drill-through.	Single-pane intelligence view for the PM
Export Engine	One-click PDF or DOCX snapshot preserving current filter state, verbatims, and metric context.	Leadership-ready intelligence brief
Insight → Action Bridge	Generates PRFAQ (PRESS RELEASE FAQ) customer problem draft, experiment hypothesis template, JIRA ticket structure per top theme.	Actionable downstream artifacts from each insight

Core Benefits

- Eliminates 3–5 days of manual synthesis per discovery cycle, down to under 60 minutes.
- Combines qualitative and quantitative signal in one scored, filterable view for the first time.
- Produces verbatim-traceable output, every theme links back to original source quotes.
- Routes themes automatically to Amazon department and product feature area.
- Generates PRFAQ (PRESS RELEASE FAQ)-ready customer problem statements directly from top-ranked themes.
- Exports leadership-ready PDF/DOCX briefs with filter state preserved.

6.1 Features Details

Feature 1: Multi-Source Data Ingestion

Qualitative Sources	Quantitative Sources	Input Format (MVP)
User interview transcripts	CSV exports from Tableau	CSV / plain text upload

Customer complaints (mail, call, chat)	Analyst SQL query outputs	CSV / Excel (.xlsx)
Support tickets	KPI dashboard exports	CSV / Excel (.xlsx)
NPS scores + open-text comments	A/B experiment results	CSV with score + comment columns
App reviews (Play Store, App Store)	Excel data from analytics team	CSV / Excel (.xlsx)
Chat logs	,	Plain text or CSV
Survey scores + segment + comments	,	CSV with segment metadata
JIRA tickets (description + status)	,	CSV export from JIRA

Feature 2: AI Processing Pipeline (6 Stages)

#	Stage	What Happens	Output
1	**Ingestion & Normalisation**	Strip formatting, handle encoding, unify schema: source type, text, date, rating, segment metadata.	Normalised JSON rows
2	**Theme Extraction**	LLM extracts the single most specific pain point per feedback unit. Returns: theme label, severity signal (H/M/L), verbatim quote.	Per-row theme JSON
3	**Semantic Clustering**	Groups similar theme labels into canonical themes. Max 8 top-level clusters. Deduplicates synonyms.	Theme taxonomy map
4	**Opportunity Scoring**	Calculates OS = (Σ severity weights) / N × 100. Weights: High=3, Medium=2, Low=1.	Ranked theme list with OS
5	**Dept + Feature Routing**	Classifies each theme by Amazon department (e.g., Outbound Logistics, Buyer Quality) and product feature area (e.g., Checkout, Search).	Routing tags per theme
6	**Quant Interpretation**	Reads analyst CSV/Excel data. Produces: metric movement summary, plain-language explanation, hypothesis implications, risk flags.	Quant context JSON per theme

Feature 3: Interactive Dashboard

- Quick-view summary cards at top: Total inputs processed, Top 5 problems ranked by OS, Average sentiment score, Risk distribution (H/M/L), Affected segments count.
- Filter criteria: Date range, Source type (qual/quant/both), Product area, Department, Sentiment level, Risk level, User segment, Minimum frequency threshold.
- Drill-through: Click any theme card → opens full detail view with verbatim clusters, metric linkage, routing, and PRFAQ (PRESS RELEASE FAQ) draft.

- Trend view: Time-series line showing theme frequency and sentiment over the selected date range.

Feature 4: Export Engine

- Filtered PDF snapshot: Preserves current filter state, shows only themes matching filters, includes verbatims and metric context.
- DOCX intelligence brief: Structured document with opportunity scorecard, theme deep-dives, metric summaries, formatted for leadership review.
- CSV export: Raw scored theme data for analysts to run further queries.

6.2 Sub Features Details

Opportunity Scoring Logic

$$\text{Opportunity Score (OS)} = \left(\Sigma\ \text{severity_weight}_i \text{ for all } i \text{ in theme} \right) \div N_{total} \times 100$$

Severity Weight Table

Signal Level	Weight	Language Indicators	Example Phrases
HIGH	3	"can't," "broken," "refund," "stopped," "returned," "won't use," "lost," "terrible"	"I can't check out." "The app is broken." "I want a refund."
MEDIUM	2	"frustrated," "slow," "confusing," "takes too long," "hard to find," "annoying"	"This is really slow." "I was confused by the address."
LOW	1	"would be nice," "prefer," "suggest," "could improve," "wish it had"	"Would be nice if I could save addresses." "Prefer a cleaner UI."

Example Output: Opportunity Scorecard

#	Theme	Mentions	% Total	OS	Sentiment	Risk	Segment
1	**Checkout Address Confusion**	482	14.3%	**18.7**	78% Neg	HIGH	Mobile Prime
2	**Delivery ETA Mismatch**	361	10.7%	**14.2**	71% Neg	HIGH	All Users
3	**Search Result Irrelevance**	294	8.7%	**10.1**	65% Neg	MEDIUM	Desktop
4	**Return Label Inaccessibility**	218	6.5%	**8.4**	82% Neg	HIGH	Mobile
5	**Slow App Load on PDP**	189	5.6%	**6.1**	58% Neg	MEDIUM	Android

6.3 Theme Deep-Dive Card (Example)

Theme: Checkout Address Confusion

Total Mentions	% of All Feedback	Opportunity Score	Risk Level
482	14.3%	18.7 / 100	HIGH

Sentiment Breakdown

Negative	Neutral	Positive
78% (376 mentions)	14% (67 mentions)	8% (39 mentions)

Recurring Phrase Clusters

- "Address keeps resetting after I edit it"
- "Cannot edit my saved address during checkout"
- "Wrong default address, shows old one every time"
- "Why does it not remember my new address?"
- "Placed order to old address, cannot change it now"

Affected Segment

Primary: Mobile Prime users (Android + iOS)
Secondary: Users with 3+ saved addresses
Geography: US and IN highest concentration

Amazon Routing

Product Area: Checkout, Address Management
Department Owner: Checkout Experience Team
Secondary Routing: Prime Identity & Account Team (address sync issue)
JIRA Label Suggestion: checkout-address-regression, mobile-prime, high-severity

Quantitative Context

Metric Signal	Checkout conversion rate dropped 2.3% in the last 4 weeks for mobile Prime users (from analyst SQL export).
Plain-Language Summary	Roughly 1 in 40 additional mobile Prime users who start checkout are not completing it compared to 4 weeks ago. The drop began in the same week the address UI was updated.
Hypothesis Implication	The address reset bug introduced in the last release is likely the primary driver. Fixing it may recover the 2.3% conversion gap.

Risk Flag	If this affects 14.3% of all feedback and correlates with a 2.3pp conversion drop, estimated revenue impact at Amazon checkout scale is material. Recommend P0 investigation.

PRFAQ (PRESS RELEASE FAQ)-Ready Customer Problem Statement

> *Mobile Prime customers who manage multiple delivery addresses cannot reliably update or select the correct address during checkout. When they attempt to edit a saved address or set a new default, the system reverts to a previously saved address, often the wrong one, causing orders to be dispatched to unintended locations. This forces customers to cancel orders, contact support, or abandon checkout entirely. The problem is concentrated in the mobile app and disproportionately affects Prime members with 3 or more saved addresses. Current workarounds require customers to delete all addresses and re-enter them, a process that erodes trust and directly harms checkout conversion.*

7. Objective

For Users

User Benefit	Current State	Target State
Synthesis speed	6–12 hours manual per cycle	Under 60 minutes from upload to scored output
Signal completeness	Qual OR quant, never both	Both combined in one filterable view per theme
Scoring objectivity	Gut-feel theme ranking	Algorithmic Opportunity Score: frequency × severity
Source traceability	Themes paraphrased from memory	Every theme links to verbatim original quotes
PRFAQ (PRESS RELEASE FAQ) readiness	Blank page, hours of writing	Draft customer problem statement auto generated
Department routing	Manual handoff decisions	AI classifies each theme to feature area and team owner
Leadership export	Copy-paste into slides/docs	One-click PDF/DOCX intelligence brief

For Business

Business Benefit	Mechanism	Expected Impact
Operational Efficiency	Replace 6–12 hrs manual synthesis with <60 min automated pipeline	Save 200+ hrs/year per PM across the org

Decision Quality	Score-backed opportunity ranking replaces intuition-driven prioritisation	Fewer Fire Phone-style misidentified opportunities
WBR (WEEKLY BUSINESS REVIEW) Preparation	Auto-generate metric narrative context linked to qual themes	Free Mon–Tue strategic bandwidth currently consumed by reporting
PRFAQ (PRESS RELEASE FAQ) Culture Fit	Generate customer problem statement grounded in scored evidence	Faster PRFAQ (PRESS RELEASE FAQ) review pass rate; fewer "weak customer problem" rejections
Escalation Intelligence	Cluster complaints rather than treating each as isolated	Detect systemic issues earlier; reduce repeat investigation cost
Brand Positioning	Demonstrates AI-native product tooling built for Amazon's workflow	Portfolio differentiation; interview-ready proof of PM systems thinking

8. Executive Summary

The Problem

Amazon Product Managers face a structural intelligence gap between the qualitative customer signals they collect (reviews, tickets, NPS comments, interview transcripts) and the quantitative business data they analyse (KPI dashboards, analyst SQL outputs, A/B experiment results). These two data types live in entirely separate systems , VoC tools for qualitative, Redshift and Tableau for quantitative, and are never synthesised together in real time. Bridging this gap manually requires 6–12 hours of effort per discovery cycle: exporting data, reading hundreds of rows, manually tagging themes, writing summary bullets, and estimating impact from memory. The result is biased, incomplete, and untraceable insight that PMs use to justify roadmap decisions worth millions of dollars.

The Proposed Solution

PM Insight Engine is a six-stage AI synthesis pipeline with an interactive dashboard front end, built for Amazon L5–L7 PMs. The PM uploads qualitative and quantitative data files (CSV/Excel). The pipeline automatically ingests, normalises, extracts themes, clusters them semantically, scores each by frequency and severity, routes them to the appropriate Amazon department and product feature area, interprets the quantitative context in plain language, and generates a PRFAQ (PRESS RELEASE FAQ)-ready customer problem statement for the top-ranked themes. The output is a unified Opportunity Dashboard: filterable, drillable, and exportable as a PDF or DOCX leadership brief. Every theme is traceable to original verbatim quotes. No insight is asserted without evidence.

Expected Outcomes

Within 90 days of pilot adoption, PM Insight Engine is expected to reduce qualitative synthesis time from 6–12 hours to under 60 minutes per discovery cycle, increase the percentage of PRFAQ (PRESS RELEASE FAQs) with evidence-backed customer problem sections from near zero to 60% of pilot team submissions, and improve PM self-rated output quality from baseline to 4.0/5.0. Structurally, it converts the PM's role in discovery from manual synthesiser to strategic reviewer, spending cognitive capacity on judgment calls, not data wrangling. The prototype is buildable in n8n and Claude using CSV inputs

within a two-week sprint, demonstrating the full pipeline before any proprietary system integration is required.

9. Feature Flow

End-to-End User Journey

The following describes how a primary user (L5/L6 Amazon PM) interacts with PM Insight Engine from data upload through to a leadership-ready export.

Step	PM Action	System Response	Time
1	Navigate to PM Insight Engine. Select "New Analysis Session."	Dashboard shows upload panel with qual and quant input zones. Displays accepted formats: CSV, Excel (.xlsx), plain text.	< 1 min
2	Upload qualitative files (reviews CSV, tickets CSV, interview transcript text).	System validates format. Confirms row count and source types detected. Flags any rows below 50 characters.	2–3 min
3	Upload quantitative files (Tableau CSV export, analyst SQL output, KPI Excel).	System validates format. Detects numeric columns. Maps to metric schema.	2–3 min
4	Set filter criteria for this session (date range, product area, segment if known).	Pre-filters applied to ingestion scope. PM sees estimated row count before processing starts.	1–2 min
5	Click "Run Analysis."	Pipeline executes all 6 stages. Progress bar shows: Normalising → Extracting → Clustering → Scoring → Routing → Generating. Estimated 3–8 min for 200 rows.	3–8 min
6	Review Opportunity Scorecard (top 5 theme cards).	Cards show theme name, Opportunity Score, risk badge (H/M/L), frequency count, sentiment %, affected segment, trend arrow.	5–10 min
7	Click into a theme card (e.g., "Checkout Address Confusion") for drill-through.	Theme Detail Panel opens verbatim clusters, sentiment breakdown, quant context panel, department routing, PRFAQ (PRESS RELEASE FAQ) draft.	5–10 min
8	Apply additional filters (e.g., Mobile only, last 30 days).	Scorecard re-ranks based on filtered dataset. All metrics update in real time.	1–2 min
9	Click "Export Intelligence Brief."	System generates PDF or DOCX with current filter state: scorecard, top 3 theme deep-dives, metric summaries, verbatims, PRFAQ (PRESS RELEASE FAQ) drafts.	< 2 min
10	Share export with cross-functional team or use PRFAQ (PRESS	PM copies PRFAQ (PRESS RELEASE FAQ) customer problem statement into their	PM-owned

	RELEASE FAQ) draft as starting point.	PRFAQ (PRESS RELEASE FAQ) document. Routing info sent to relevant team.	

Sub-Feature 1: Data Ingestion & Normalisation Engine

User Story

> *"As an Amazon PM, I want to upload my qualitative and quantitative data files in the formats I already use (CSV, Excel, plain text), so I do not have to reformat my data before the tool can process it."*

Flow

1. PM drags and drops files onto the upload panel or clicks to browse.
2. System detects file type and source type from filename or PM-selected dropdown (reviews / tickets / NPS / interviews / analyst data / KPI export).
3. System reads first 5 rows and displays a preview with detected column mapping.
4. PM confirms or corrects the mapping (e.g., "which column is the feedback text?").
5. System applies minimum length filter (50 chars). Rows below threshold are flagged in a "Skipped rows" log accessible to the PM.
6. System normalises all accepted rows to unified schema: {source type, text, date, rating, segment, is qualitative: bool}.

Visual Flow

Upload Panel → Format Validation → Column Mapping Preview → PM Confirmation → Normalisation → Unified Row Store → Proceed to AI Pipeline

Edge Cases

- Empty file uploaded: System shows error "File contains no readable rows. Please check formatting and retry."
- Mixed-language reviews: System accepts and processes. Language is detected per row. Non-English rows are flagged in the output with language tag.
- Duplicate file uploaded: System detects identical filename + row count. Warns: "This file appears to have been uploaded already. Proceed or deduplicate?"
- No date column present: System uses upload date as fallback. Informs PM: "No date column detected , using today's date for trend analysis."
- Excel with multiple sheets: System displays sheet selector. PM chooses which sheet(s) to include.

Sub-Feature 2: AI Synthesis Pipeline (6 Stages)

User Story

> *"As an Amazon PM, I want the tool to automatically extract themes, score them, and route them, so I do not have to manually tag, rank, or triage feedback across hundreds of rows."*

Flow , 6 Pipeline Stages

Stage	Name	AI or Logic	Input	Output	Failure Mode
1	Theme Extraction	Gemini 1.5 Pro	Normalised rows (batches of 50)	{theme label, severity: H/M/L, verbatim quote} per row	Row has no extractable theme → flagged as "unclassified," excluded from scoring
2	Semantic Clustering	Gemini 1.5 Pro	All extracted theme labels	Canonical theme map: {canonical theme: [input labels]}. Max 8 themes.	Too many unique labels → clustering prompt retried with stricter grouping instruction
3	Opportunity Scoring	n8n logic (deterministic)	Clustered themes + severity weights	OS = (Σ severity weights / N_total) × 100. Risk: >15=HIGH, 8–15=MED, <8=LOW	Zero-count theme after clustering → automatically suppressed from output
4	Department Routing	Gemini 1.5 Pro	Top 8 canonical themes	{feature area, department owner} per theme	Low-confidence routing → flagged as "Suggested , verify with team." PM can override.
5	Quant Interpretation	Gemini 1.5 Pro	Quantitative CSV/Excel + top themes	{metric summary, hypothesis, risk flag} per theme	No quant data uploaded → quant panel shows "No quantitative data linked for this session."
6	PRFAQ (PRESS RELEASE FAQ) Generation	Gemini 1.5 Pro	Top 3 themes + their evidence	4–5 sentence customer problem statement in PRFAQ (PRESS RELEASE FAQ) style	Generated draft too generic → PM can regenerate with a "Try again" button that adds more specific context

Edge Cases

- All feedback is 1-star reviews: System warns in dashboard header: "Input skewed: 92% of rows are 1–2-star reviews. Scores reflect this distribution." Scores are still valid within that dataset.
- Fewer than 20 rows ingested: System warns that results may not be statistically representative. Proceeds but adds a "Low sample size" badge to all theme cards.
- AI API timeout during processing: Stage is retried up to 3 times. If all retries fail, PM is notified and can resume from the failed stage without re-uploading.

- Quant data contains no numeric columns: System flags the file as text-only and routes it through the qualitative pipeline instead.

Sub-Feature 3: Unified Opportunity Dashboard

User Story

> *"As an Amazon PM, I want a single interactive dashboard that shows my top customer problems ranked by evidence , with filters I can adjust and drill-through to verbatims, so I can prepare for a leadership review without opening five different tools."*

Flow

7. PM views the Opportunity Dashboard after pipeline completes.
8. Top of screen: 4 quick-view summary cards, Total Inputs Processed, Top Problem Theme, Average Sentiment Score, Risk Distribution (H/M/L count).
9. Main panel: Opportunity Scorecard table showing top 5 themes ranked by OS. Each row shows: rank, theme name, frequency count, % of total, OS, sentiment, risk badge, affected segment, trend arrow.
10. PM clicks column headers to re-sort by any field (e.g., sort by sentiment % to see most negative themes first).
11. PM applies filters from the left panel: date range, source type, product area, department, risk level, segment, minimum frequency.
12. Scorecard updates in real time based on filter selection.
13. PM clicks a theme row → Theme Detail Panel opens in right panel or full-screen modal.
14. Theme Detail Panel shows sentiment breakdown donut, verbatim clusters (top 5 phrases), quant context panel, routing tags, PRFAQ (PRESS RELEASE FAQ) draft with copy button.

Visual Flow

Summary Cards → Scorecard Table (filterable, sortable) → Click Theme → Theme Detail Panel (verbatims + quant + routing + PRFAQ (PRESS RELEASE FAQ) draft)

Edge Cases

- No themes scored above OS 5.0: Dashboard shows all themes but flags: "Low signal density. Consider expanding date range or adding more data sources."
- Filter results in zero themes: Dashboard shows empty state with message: "No themes match current filters. Adjust filter criteria to see results."
- Trend data unavailable (no date column): Trend arrows are hidden. A tooltip explains: "Trend view unavailable, no date data in source files."
- PM sorts by a column while a filter is active: Sort applies within the current filtered dataset, not the full dataset.

Sub-Feature 4: Theme Deep-Dive & Drill-Through

User Story

> *"As an Amazon PM, I want to click into any theme and see the full evidence: verbatim quotes, sentiment breakdown, which customers are affected, what the metrics say, and a draft of the customer problem , so I have everything I need to write a PRFAQ (PRESS RELEASE FAQ) or brief a team lead."*

Flow

15. PM clicks on a theme in the Opportunity Scorecard.
16. Theme Detail Panel opens. Header shows: theme name, OS, risk badge, frequency count, % of total, affected segment.
17. Sentiment Breakdown section: donut chart showing % negative / neutral / positive. Clicking a segment filters verbatims to that sentiment only.
18. Verbatim Clusters section: top 5 recurring phrase clusters with example quotes. Each quote shows source type and date. PM can click "Show more" to see up to 20 verbatims.
19. Quantitative Context panel: metric movement summary, plain-language explanation, hypothesis implication, risk flags, all linked to the theme.
20. Routing section: feature area, department owner, suggested JIRA label. PM can override routing via dropdown. Override is saved for the session.
21. PRFAQ (PRESS RELEASE FAQ) Draft section: 4–5 sentence customer problem statement. Copy button copies to clipboard. "Regenerate" button prompts AI for an alternative draft.
22. Experiment Hypothesis Template: pre-filled hypothesis statement based on theme data. PM can edit inline.

Edge Cases

- Theme has fewer than 3 verbatims: All available verbatims are shown. A note reads: "Low verbatim count for this theme, treat PRFAQ (PRESS RELEASE FAQ) draft as indicative only."
- No quantitative data linked: Quant Context panel shows: "No quantitative data was uploaded for this session. Upload an analyst report or KPI export to see metric context."
- PM overrides routing: Override is saved with a "PM-confirmed" badge. Original AI suggestion remains visible for comparison.
- PRFAQ (PRESS RELEASE FAQ) draft is too generic: PM clicks "Regenerate." System adds current verbatims and segment data as additional context in the prompt and produces a new draft.

Sub-Feature 5: Export Engine

User Story

> *"As an Amazon PM, I want to export a clean, formatted intelligence brief that reflects exactly what I see in the filtered dashboard, so I can share it with my director or cross-functional team without reformatting anything."*

Flow

23. PM clicks "Export Intelligence Brief" button in the top right of the dashboard.
24. Export options modal appears: PDF (recommended for sharing) or DOCX (recommended for editing). PM selects format.

25. PM optionally selects which themes to include (default: all currently visible after filters).
26. System generates export document containing session metadata (date, PM name, data sources, row count), summary cards, filtered Opportunity Scorecard, top 3 theme deep-dives with verbatims and quant context, PRFAQ (PRESS RELEASE FAQ) customer problem drafts, routing table, risk summary.
27. Download begins automatically. File is named: "Insight_Brief_[Product Area] _[Date].pdf" or ".docx."

Edge Cases

- PM exports with no filters applied: Export includes all themes. A note at the top reads: "No filters applied, this export reflects the full dataset."
- PM exports with only 1 theme visible after filtering: Export is generated for that single theme only. All sections reflect that theme's data.
- Export file exceeds 10MB (very large dataset): System generates a summary-only version and notifies PM. Full verbatim export available as separate CSV download.

10. Edge Cases

The following consolidates all edge cases across sub-features with proposed system handling for each.

Sub-Feature	Edge Case	System Handling	Priority
Ingestion	File is empty or unreadable	Error message with format guide. No processing attempted.	HIGH
Ingestion	No date column in file	Upload date used as fallback. PM notified via info banner.	MEDIUM
Ingestion	Mixed languages in feedback	Rows processed; language detected per row. Non-English rows tagged in output.	MEDIUM
Ingestion	Duplicate file uploaded	Duplicate detected by filename + row count. PM prompted to deduplicate or proceed.	MEDIUM
Ingestion	Excel with multiple sheets	Sheet selector displayed. PM chooses which sheets to include.	LOW
AI Pipeline	All input is 1-star reviews	Processing continues. Warning banner: "Input skewed, scores reflect this distribution."	HIGH
AI Pipeline	Fewer than 20 rows	"Low sample size" badge on all cards. PM warned results are indicative only.	MEDIUM
AI Pipeline	API timeout during a stage	Auto-retry x3. If all fail, PM notified. Resume from failed stage without re-upload.	HIGH

AI Pipeline	Quant file has no numeric columns	File routed through qual pipeline. PM notified via info banner.	MEDIUM
AI Pipeline	Theme extraction finds no themes in a row	Row flagged as "unclassified." Excluded from scoring. Visible in processing log.	LOW
Dashboard	Filters result in zero themes	Empty state message: "No themes match filters. Adjust criteria."	HIGH
Dashboard	No themes above OS 5.0	All themes shown. Low signal density warning displayed.	MEDIUM
Dashboard	PM applies filter while sorting	Sort applies within the filtered dataset. Sort state preserved when filter is removed.	LOW
Drill-Through	Theme has < 3 verbatims	All verbatims shown. Note: "Low verbatim count, treat PRFAQ (PRESS RELEASE FAQ) draft as indicative."	MEDIUM
Drill-Through	No quant data uploaded	Quant panel shows placeholder with upload prompt.	LOW
Drill-Through	PRFAQ (PRESS RELEASE FAQ) draft is too generic	PM clicks "Regenerate." Prompt enriched with verbatims and segment. New draft generated.	MEDIUM
Drill-Through	PM overrides routing	Override saved with "PM-confirmed" badge. Original AI suggestion remains visible.	LOW
Export	PM exports with no filters	Full dataset exported. Note added at top of document.	LOW
Export	Export file > 10MB	Summary-only version generated. Full verbatim export offered as separate CSV.	MEDIUM
Export	Single theme visible at export time	Single-theme export generated normally. All sections reflect that theme.	LOW

11. Success Metrics

Main Metrics (Core KPIs)

KPI	Definition	Baseline	Target (90 days)
Synthesis Time	Hours from raw data upload to scored intelligence brief available in dashboard	6–12 hrs manual	< 60 minutes

PRFAQ (PRESS RELEASE FAQ) Adoption Rate	% of PRFAQ (PRESS RELEASE FAQs) where Insight Engine output is cited in customer problem section	0%	≥ 60% of pilot PRFAQ (PRESS RELEASE FAQs)
Theme Traceability Rate	% of ranked themes with ≥ 3 verbatim quotes in drill-through view	N/A	≥ 95% of themes
Decision Velocity	Days from feedback collection to first PRFAQ (PRESS RELEASE FAQ) draft submitted	5–7 days avg	≤ 2 days

Adoption Metrics

Stage	Metric	Definition	Target
Awareness	Pilot PM onboarding rate	% of pilot team PMs who complete first upload within 2 weeks of launch	100% of pilot cohort
Activation	First export rate	% of PMs who export a PDF/DOCX brief within their first session	≥ 50%
Usage	Uploads per PM per month	Average number of data synthesis sessions per PM per 30 days	≥ 3 sessions/month
Engagement	Drill-through rate	% of dashboard sessions where PM clicks into at least 1 theme detail card	≥ 70% of sessions
Retention	Month 2 return rate	% of pilot PMs who use the tool at least once in Month 2	≥ 80%

Impact Metrics

Metric	Definition	Target
PRFAQ (PRESS RELEASE FAQ) pass rate	% of PRFAQ (PRESS RELEASE FAQs) that pass leadership review on first submission (customer problem section)	Improve by +15pp vs. pre-tool baseline
Escalation re-investigation rate	% of escalations where root cause was already clustered in a prior synthesis session	Reduce by 40% within 90 days

Opportunity identification lag	Days between a customer problem emerging in feedback and being surfaced as a ranked theme	Reduce from ~14 days to <48 hours
Cross-source theme match rate	% of top 5 themes that appear in both qual and quant inputs simultaneously	Baseline measurement in Month 1

Quality Metrics

Metric	Definition	Target
PM output quality rating	PM self-rates output specificity vs. manual synthesis on 1–5 scale (post-session survey)	≥ 4.0 / 5.0
PRFAQ (PRESS RELEASE FAQ) draft usefulness	% of generated PRFAQ (PRESS RELEASE FAQ) customer problem drafts used as a starting point (vs. discarded)	≥ 65% used as draft base
Routing accuracy	% of AI-suggested department routings confirmed as correct by PM (not manually overridden)	≥ 80% confirmed correct
Verbatim relevance	PM rating of verbatim quote relevance to the identified theme (1–5 survey item)	≥ 4.2 / 5.0

Business Metrics

Metric	Definition	Target
PM hours saved per week	Reduction in self-reported synthesis time per PM per week	≥ 5 hours/week/PM
WBR (WEEKLY BUSINESS REVIEW) prep time reduction	Self-reported hours spent on Monday/Tuesday WBR (WEEKLY BUSINESS REVIEW) prep before vs. after tool adoption	Reduce by ≥ 30%
Cross-team referral rate	% of pilot PMs who recommend the tool to another PM unprompted	≥ 40% by Month 2
Export utilisation	% of synthesis sessions that result in a PDF or DOCX export	≥ 40% of sessions

Diagnostic Framework

If Synthesis Time Is Not Improving

28. Check upload success rate, are PMs failing to format CSVs correctly? Add a template and validation error messages.
29. Check pipeline processing time, is the AI API latency the bottleneck? Batch size may need reduction.
30. Interview 3 PMs, are they re-doing manual synthesis in parallel because they don't trust the output? Trust issue, not speed issue.

If PRFAQ (PRESS RELEASE FAQ) Adoption Rate Is Low

31. Review 5 generated PRFAQ (PRESS RELEASE FAQ) drafts, are they too generic? Prompt engineering needs tightening with more domain-specific examples.
32. Check if PMs are reading the draft, add a copy-to-clipboard button directly on the theme card to reduce friction.
33. Survey non-adopting PMs, is it a quality issue or a workflow habit issue? Different interventions for each.

If Routing Accuracy Is Below 80%

34. Audit 20 misrouted themes, is the error concentrated in specific theme types (e.g., ambiguous complaints) or spread evenly?
35. Add more specific routing examples to the AI prompt. Consider a PM override + feedback loop to improve routing classification over time.

10. Open Questions / Collaboration Notes

How to Use This Section

This section is a live working space for product, engineering, design, and business stakeholders to log questions, decisions needed, assumptions to validate, and feedback on any section of this PRD. Add your name, the date, and the section number your comment refers to.

Open Questions , Product

#	Question	Owner	Status	Notes
P1	Should the Opportunity Score formula weight severity vs. frequency differently for escalation-sourced data (e.g., escalations are intrinsically high-severity)?	Product	Open	Consider a source-type multiplier in v1.1
P2	Should the dashboard support multi-session comparison (compare this week's scorecard to last week's)?	Product	Open	Requires persistent session storage, likely Phase 2
P3	Should the PRFAQ (PRESS RELEASE FAQ) draft be shown for all 5 themes or only the top 3?	Product	Open	Top 3 preferred for MVP to keep generation cost and latency low
P4	How should the tool handle confidential or PII-containing feedback? (e.g., customer names in reviews)	Product / Legal	Needs Legal Input	PII detection and redaction may be required before AI processing
P5	Should the tool support collaborative annotation (multiple PMs commenting on the same session)?	Product	Deferred to Phase 2	Solo PM workflow validated first before team features added

Open Questions , Engineering

#	Question	Owner	Status	Notes
E1	What is the Gemini 1.5 Pro API rate limit for the batch extraction step? How many parallel calls can we make for a 500-row input?	Engineering	Open	Affects processing time estimate and batch size decisions
E2	Should n8n be self-hosted or cloud for the prototype? Which has lower latency for the pipeline?	Engineering	Open	Cloud n8n preferred for portfolio demo accessibility

E3	Can Looker Studio connect directly to n8n JSON output, or do we need an intermediate Google Sheet?	Engineering	Open	Google Sheet intermediate is simplest for prototype
E4	What is the best export format for the DOCX export, Google Docs API or a docx library (python-docx)?	Engineering	Open	Google Docs API preferred for richer formatting
E5	How do we handle Excel files with merged cells or non-standard headers in the ingestion normalisation step?	Engineering	Open	Needs a pre-processing step to flatten Excel structure before column mapping

Open Questions , Design

#	Question	Owner	Status	Notes
D1	Should the Theme Detail Panel open as a side panel or a full-screen modal? Which maintains context better for the PM?	Design	Open	Side panel preferred to keep scorecard visible for comparison
D2	What is the most effective way to visualise the verbatim clusters, word cloud, grouped list, or expandable quote cards?	Design	Open	Expandable quote cards preferred, word clouds lack traceability
D3	How do we make the risk badge (HIGH/MEDIUM/LOW) accessible for colour-blind users while maintaining visual distinction?	Design	Open	Add icon + text label in addition to colour coding
D4	Should the upload panel support drag-and-drop from tools like Tableau or is file upload sufficient for MVP?	Design	Open	File upload sufficient for prototype. Tableau integration is Phase 2.

Assumptions to Validate

Assumption	How to Validate	Risk if Wrong
PMs will adopt a CSV upload workflow without native Redshift/Tableau integration	User interviews with 3–5 L5/L6 PMs before build. Ask about current export habits.	Tool adoption is low because friction of exporting to CSV is a barrier. Solution: add Tableau CSV export guide in onboarding.
Gemini 1.5 Pro extraction quality is sufficient for Amazon-domain feedback	Run extraction on 200 real Amazon review rows. Grade output vs. manual tagging. Target: ≥ 85% theme label match.	Extraction quality is too low for PM trust. Solution: add few-shot examples of Amazon-specific themes to extraction prompt.

PMs will trust AI-generated PRFAQ (PRESS RELEASE FAQ) drafts as a starting point (not replace them)	Survey pilot PMs after first session. Ask: "Did you use any part of the PRFAQ (PRESS RELEASE FAQ) draft?"	PMs ignore the draft entirely. Solution: reframe as "Evidence summary" rather than "PRFAQ (PRESS RELEASE FAQ) draft", less prescriptive framing.
Department routing will have >80% accuracy for common Amazon team structures	Test routing against 50 known themes with ground-truth team assignments.	Routing is inaccurate and erodes trust. Solution: add a confidence score to each routing suggestion. Low-confidence routings shown with a prominent "verify" flag.

Add Your Feedback Here

> *Instructions: Add your name, date, section reference, and comment below. Use [QUESTION], [DECISION NEEDED], [ASSUMPTION], or [FEEDBACK] tags to categorise your input. Example: "[QUESTION] P1, John S., 15 Mar 2026 , Should we also include JIRA ticket data as a qualitative input source in MVP?"*

[Add feedback here]